# Bridging Text and Graph Topology for Wikipedia Article Recommendation

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**Abstract.** Article recommendation systems identify relevant articles based on reader preferences and reading history. Existing approaches include collaborative filtering, matrix factorization, content-based methods, and neural network-based models. Recently, Graph Neural Networks (GNNs) have become central to article recommendation by leveraging article network structure. These systems are often evaluated on benchmarks such as Cora, CiteSeer, PubMed, and Wiki-CS. Among them, Wiki-CS is widely used due to its open-source accessibility, graph-ready structure, and representation of Wikipedia articles as nodes with directed hyperlink edges. Article text can also be embedded and used as node attributes, enabling richer representations.
This study presents a graph-based article-to-article recommendation framework centered on hybrid semantic-structural retrieval. We combine sentence embeddings with structural representations from Node2Vec, and study corrected Node2Vec variants together with hybrid models that fuse text and graph structure. We also investigate pooled semantic representations designed to balance textual and structural signals, yielding corrected hybrid models with the most balanced recommendation quality in our experiments. We benchmark semantic, structural, hybrid, and GNN-based models under a revised evaluation protocol, showing that carefully designed hybrid retrieval is more reliable than single-source baselines.
To support this setting, we introduce a reconstructed and up-to-date version of Wiki-CS. The reconstruction resolves the lack of usable raw text in the original benchmark by cleaning and aligning article titles, recovering article text through integration with the Wikipedia STEM corpus and Wikipedia API, and applying state-of-the-art embedding models to the reconstructed collection. This provides complete raw text for every active node and enables reproducible, up-to-date recommendation experiments with modern Transformer-based text representations.

**Keywords:** Article recommendation · Graph recommendation · Hybrid retrieval · Wikipedia graphs · Benchmark reconstruction

## 1 Introduction

Article recommendation can be studied under several settings. Many widely used systems rely on user behavior, collaborative signals, or historical interaction pat-

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terns. The setting considered in this paper is different: given a source Wikipedia article, the objective is to retrieve a ranked list of related articles that are both semantically relevant and structurally meaningful in the hyperlink graph. This article-to-article formulation is useful when explicit user histories are unavailable, when interpretability matters, or when the goal is guided exploration of a knowledge domain rather than personalized consumption.

Wikipedia is a natural graph environment for this task. Its hyperlink structure encodes editorial and conceptual relations, while article text provides a dense semantic signal that can reveal topical proximity even when direct links are missing. These two sources of information are complementary, but they are not always easy to combine in a way that preserves both topological context and textual meaning. This makes Wikipedia article recommendation a suitable testbed for hybrid semantic-structural retrieval.

The starting point of this work is Wiki-CS, a widely used benchmark derived from Wikipedia and originally designed for node classification experiments [11]. Although Wiki-CS offers a graph-ready representation and broad accessibility, it does not directly provide a text-complete, up-to-date benchmark for article recommendation with modern text encoders. This creates two related needs: an updated dataset layer with recoverable article text, and a recommendation framework able to compare semantic, structural, hybrid, and GNN-based approaches under a shared evaluation setting.

This paper addresses both needs. First, we reconstruct Wiki-CS into a text-enriched benchmark, Custom-WikiCS, by aligning titles, recovering article text, and attaching modern embedding representations. Second, we introduce a graph-based article recommendation framework built around semantic retrieval, Node2Vec-based structural retrieval, corrected structural variants, pooled semantic-structural hybrids, and graph-neural baselines. Third, we evaluate these model families under a revised undirected protocol that is better aligned with the structural view of the task.

The main contributions of the paper are:

- a reconstructed version of Wiki-CS, denoted Custom-WikiCS, with complete raw text for every active node and support for modern text embedding pipelines;
- a hybrid recommendation framework that combines sentence embeddings with Node2Vec-derived structural signals, including corrected structural variants and pooled semantic fusion;
- a comparative benchmark across semantic, structural, hybrid, and GNN baselines under a revised evaluation protocol for article-to-article recommendation.

## 2 Related Work

Wikipedia can be modeled naturally as a graph whose nodes are articles and whose edges are hyperlinks. This makes it a practical environment for graph representation learning and graph-based recommendation. Wiki-CS provides one of the most widely reused Wikipedia-derived graph benchmarks in graph learning research [11], although its original use case is node classification rather than article recommendation. More directly related work includes WikiRecNet, which studies large-scale Wikipedia recommendation using graph structure and representation learning [12]. One lesson from this line of work is that structural proximity is informative, but recommendation quality should not be reduced to a generic graph prediction task.

Textual embeddings provide a second major signal family. Transformer-based encoders [15] and BERT-style models [2] have made dense semantic retrieval practical at sentence and document level. Sentence-BERT [13] further showed how BERT-like architectures can be adapted for efficient similarity search. More recent retrieval-oriented models, such as BGE-M3 [1], are explicitly designed for strong dense retrieval across multilingual settings and thus fit the recommendation scenario considered here.

Structural embeddings capture graph position, local neighborhood, and community patterns directly from the edge structure. Node2Vec [3] is one of the most influential methods in this family, learning node representations from random-walk contexts. In recommendation settings, structural embeddings can recover relations that text alone may miss, although they can also become sensitive to hub effects and graph-specific artifacts if the evaluation protocol is not carefully aligned with the structural task.

Graph neural networks provide another structural modeling family. GCN remains a standard transductive baseline [6], while GraphSAGE [4] introduces neighborhood aggregation with an inductive flavor. Survey work on GNNs for text-related settings suggests that these models remain central at the interface of graph structure and linguistic representation [17]. In parallel, multimodal and hybrid recommender systems increasingly combine text and graph signals rather than relying on either source alone [8]. Practical hybrid examples based on BERT and GNN components also show that fusion design can change retrieval behavior substantially [5].

The present work is positioned at the intersection of these directions. It uses Wikipedia as a graph recommendation environment, updates a widely used benchmark to support modern text modeling, and compares semantic, structural, hybrid, and GNN-based article recommendation models within one experimental framework.

## 3 Method

### 3.1 Custom-WikiCS Reconstruction

**Motivation and Reconstruction Pipeline** The original Wiki-CS graph is useful as a structural benchmark, but it is less convenient for modern article recommendation because the benchmark is graph-ready without being text-complete for current retrieval pipelines. In particular, the node set is easy to load as a graph, yet the benchmark does not directly expose the reconstructed raw article text needed to regenerate semantic features with modern encoders. For a recommendation setting centered on article-to-article retrieval, this is a practical limitation: text quality and text availability are not auxiliary metadata, but part of the representation being evaluated.

Custom-WikiCS addresses this limitation through a reconstruction pipeline with three main stages. First, article titles are cleaned and aligned in order to improve consistency across graph records, the Wikipedia STEM corpus[1], and API lookups. This stage resolves naming mismatches, formatting inconsistencies, and stale title variants that would otherwise prevent article recovery. Second, raw article text is recovered by combining the Wikipedia STEM corpus with Wikipedia API backfilling, producing a text-complete collection for the active article set. Third, the reconstructed collection is used to generate modern text representations, enabling semantic retrieval models and hybrid recommendation variants on top of the same graph. In the final reconstructed benchmark, every active node is associated with usable article text and can therefore participate in both structural and semantic evaluation.

After reconstruction and cleaning, isolated nodes are removed and the final graph contains the statistics reported in Table 1. The resulting benchmark supports both graph-structural experiments and modern text-based retrieval on the same article set.

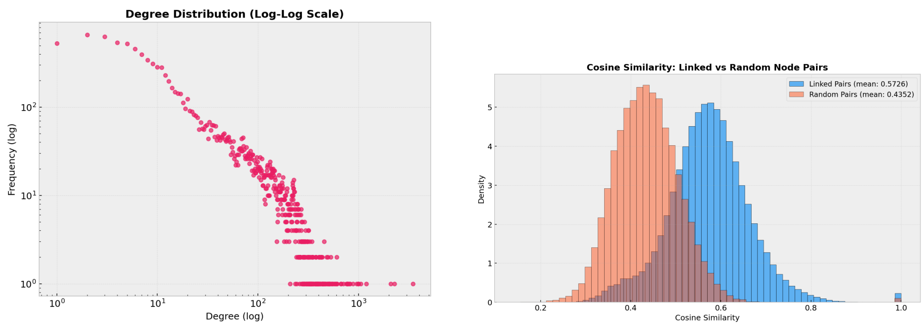
**Table 1.** Custom-WikiCS graph statistics after reconstruction and cleaning

| **Statistic** | **Value** |
|---|---|
| Initial nodes | 11,701 |
| Removed isolates | 334 |
| Active nodes | 11,367 |
| Directed edges | 291,039 |
| Undirected edges | 216,123 |

**Graph Characteristics** The cleaned graph exhibits a heavy-tailed degree distribution, indicating that a small number of hub articles coexist with a much larger set of lower-degree nodes. This matters for article recommendation because purely structural methods can become dominated by hubs if degree effects are left untreated. The degree statistics are also strongly skewed in the cleaned graph: total degree ranges from 1 to 3,474, with mean 51.21 and median 16. This gap between mean and median is consistent with a hub-dominated article network rather than a uniformly mixed graph. The left panel of Fig. 1 visualizes the same behavior on log-log axes and supports the power-law-style correction heuristic introduced later in the recommendation framework.

At the same time, the graph is semantically assortative. Linked article pairs are substantially closer in the semantic embedding space than randomly sampled

[1] `https://www.kaggle.com/datasets/conjuring92/wiki-stem-corpus`




(a) Log-log degree distribution (b) Linked vs. random cosine similarity

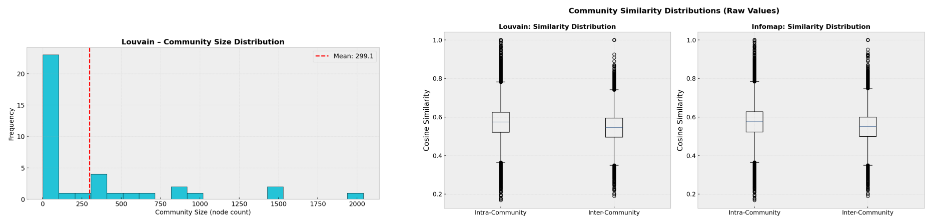
**Fig. 1.** Structural and semantic diagnostics of Custom-WikiCS. The degree profile is heavy-tailed, while linked article pairs are semantically closer than random pairs in the BGE-M3 embedding space

pairs: under cosine similarity, the linked mean is 0.5726 while the random mean is 0.4352. This is important because it shows that hyperlink connectivity and textual meaning are aligned rather than independent. In other words, semantic retrieval is not being imposed on an unrelated graph; it operates on a benchmark where the link structure already carries topic-level coherence. The right panel of Fig. 1 summarizes this effect.

Community analysis further shows that the graph is neither flat nor uniformly mixed. Louvain detection yields modularity 0.6476 over 38 communities, with a largest community of 2,040 nodes and mean community size 299.13. This confirms that the cleaned graph contains broad topic regions rather than a single homogeneous article pool. Moreover, the semantic coherence of these communities is measurable: intra-community cosine similarity is higher than inter-community similarity under both Louvain (0.5729 vs. 0.5435) and Infomap (0.5748 vs. 0.5480). This means that graph communities are not only topological partitions but also semantically coherent article groupings. Such a structure is useful for recommendation because it creates meaningful local neighborhoods beyond direct edge presence. Figure 2 summarizes the community-size distribution together with the intra/inter community similarity comparison.

### 3.2 Recommendation Framework

**Task Definition** Let $G = (V, E)$ denote the reconstructed article graph. For a source article $u \in V$, the goal is to return a ranked list of candidate articles $v \in V \setminus \{u\}$ that are relevant to $u$ under semantic, structural, or hybrid similarity. The task is query-based and article-centric: the source article serves as the query, and the recommendation model ranks all remaining articles accordingly.




(a) Louvain community size distribution

(b) Intra- vs. inter-community similarity

**Fig. 2.** Community diagnostics for Custom-WikiCS. The cleaned graph contains non-trivial communities, and those communities remain semantically coherent

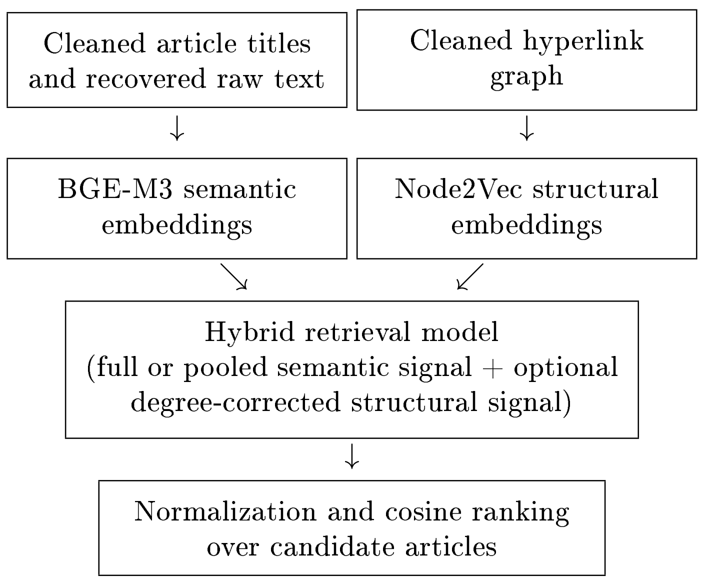



**Fig. 3.** Flagship hybrid recommendation pipeline used in the paper

**Semantic Retrieval** The semantic component uses BGE-M3 text embeddings [1]. Each article is represented by a 1024-dimensional dense vector derived from its reconstructed text. Recommendation scores in the semantic-only setting are computed by cosine similarity between source and candidate embeddings:

$$s_{\text{sem}}(u, v) = \cos(\mathbf{b}_u, \mathbf{b}_v). \tag{1}$$

This component captures semantic proximity even when the hyperlink relation is sparse or absent, and it serves as both a standalone baseline and one component of the hybrid models.

**Structural Retrieval** The structural component uses Node2Vec-style representations [3] computed on the Wikipedia hyperlink graph. Two structural variants are considered. The first is an uncorrected Node2Vec configuration, which ranks candidates using cosine similarity in the structural embedding space:

$$s_{\mathrm{str}}(u, v) = \cos(\mathbf{n}_u, \mathbf{n}_v). \tag{2}$$

The second is a corrected variant that applies a power-law degree penalty during structural scoring in order to reduce the tendency of high-degree hubs to dominate retrieval.

Following the project reports, the correction factor for a node $v$ with undirected degree $d_v$ is defined as

$$p_v = \log(d_v + 1)^{\alpha}, \tag{3}$$

where the exponent is estimated from the structural training graph and is approximately $\alpha \approx 2.9966$ in the revised evaluation setting. The logarithm softens the extreme upper tail of the degree distribution, while the power exponent preserves a stronger penalty for structurally over-connected nodes. The corrected structural score becomes

$$s_{\mathrm{corr}}(u, v) = \frac{\cos(\mathbf{n}_u, \mathbf{n}_v)}{p_u p_v}. \tag{4}$$

**Hybrid and GNN-Based Models** The central model family of this paper combines semantic and structural representations. We evaluate four main hybrid variants:

- direct concatenation of BGE-M3 and uncorrected Node2Vec;
- direct concatenation of BGE-M3 and corrected Node2Vec;
- pooled BGE-M3 with uncorrected Node2Vec;
- pooled BGE-M3 with corrected Node2Vec.

In the pooled variants, the 1024-dimensional semantic representation is reduced to 128 dimensions by block-wise pooling before concatenation, making the semantic and structural components more comparable in scale. This produces a 256-dimensional hybrid representation rather than the 1152-dimensional vector obtained by direct concatenation.

More explicitly, the non-pooled hybrid representation is formed as

$$\mathbf{h}_u = [\mathbf{b}_u; \mathbf{n}_u], \tag{5}$$

while the pooled representation uses a reduced semantic vector $\bar{\mathbf{b}}_u \in \mathbb{R}^{128}$ obtained from contiguous blocks of the original BGE-M3 embedding:

$$\mathbf{h}_u^{\mathrm{pool}} = [\bar{\mathbf{b}}_u; \mathbf{n}_u]. \tag{6}$$

In corrected hybrids, the same degree-normalized structural signal is used before concatenation. After construction, the hybrid vectors are normalized and ranked by cosine similarity against all candidate articles. The purpose of pooling is not to create a new semantic encoder, but to prevent the 1024-dimensional text representation from numerically overwhelming the 128-dimensional structural contribution.

The benchmark also includes GCN and GraphSAGE structural baselines, as well as a ranking-level fusion baseline combining GCN and BGE-M3. Together, these models span semantic-only, structural-only, hybrid-concatenative, rank-fusion, and GNN-based retrieval families.

## 4 Experiments and Results

All main results reported in this paper use the revised undirected structural evaluation protocol. Starting from the undirected version of the cleaned graph, we construct a split with 216,123 undirected edges, of which 172,900 are used for training and 43,223 for testing. To align ranking evaluation with article-to-article retrieval, each undirected test edge $\{u, v\}$ is expanded into two directed queries, $u \to v$ and $v \to u$, yielding 86,446 evaluation pairs over 9,022 evaluation nodes.

This revised protocol plays an important methodological role. Earlier project stages also reported results on held-out directed edges, but the later comparison in this paper is intentionally centered on the undirected structural graph because Node2Vec neighborhoods, community analysis, and the degree correction heuristic are all defined over that structural view. The revised protocol therefore preserves the article-to-article ranking task while aligning the training structure, the correction heuristic, and the evaluation target more closely than a purely directed holdout would.

The experimental workflow is notebook-centered and relies on cached structural and semantic artifacts generated from the reconstructed dataset. The benchmark includes semantic, structural, hybrid, and GNN-based models evaluated under the same underlying revised split, making the reported comparisons directly comparable at the level of ranking outputs.

### 4.1 Evaluation Metrics

Let $Q$ denote the set of evaluation queries, let $r_q$ be the rank of the held-out target article for query $q$, and let $L_q$ be the returned recommendation list. Our ranking metrics follow standard information retrieval practice [9, 16]. Because each query has exactly one held-out target in the ranking protocol, several metrics simplify. The basic top-list metrics are

$$\text{Hit@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[r_q \leq K] \tag{7}$$

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{r_q} \tag{8}$$

For the same single-target setting, normalized discounted cumulative gain reduces to

$$\mathrm{NDCG@}K = \frac{1}{|Q|} \sum_{q \in Q} \frac{\mathbf{1}[r_q \leq K]}{\log_2(r_q + 1)} \tag{9}$$

since the ideal DCG of one relevant item equals 1. Likewise,

$$\mathrm{Recall@}K = \mathrm{Hit@}K \tag{10}$$

and average precision becomes

$$\mathrm{AP@}K(q) = \begin{cases} 1/r_q, & r_q \leq K, \\ 0, & r_q > K, \end{cases} \tag{11}$$

$$\mathrm{MAP@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{AP@}K(q) \tag{12}$$

In the main results table, the unlabeled NDCG, Recall, and MAP columns correspond to $NDCG@100$, $Recall@100$, and $MAP@100$. Therefore Recall is numerically identical to Hit@100 in our setup.

For the discrimination-oriented extension, ROC-AUC and PR-AUC are computed from model scores assigned to positive undirected test edges and sampled negative edges. ROC-AUC measures the area traced by true-positive rate and false-positive rate across thresholds, while PR-AUC measures the area under the precision-recall curve and is especially informative under class imbalance [14]:

$$\begin{aligned} \mathrm{TPR}(\tau) &= \frac{TP(\tau)}{TP(\tau) + FN(\tau)}, \\ \mathrm{FPR}(\tau) &= \frac{FP(\tau)}{FP(\tau) + TN(\tau)}, \\ \mathrm{Precision}(\tau) &= \frac{TP(\tau)}{TP(\tau) + FP(\tau)}. \end{aligned} \tag{13}$$

We also report recommendation-quality metrics that characterize broader system behavior beyond target ranking. Let $C$ denote the candidate article set, let $d(v)$ be the undirected degree of article $v$, and let $M = \sum_{v \in C}(d(v) + 1)$. Popularity Lift is computed as

$$\text{Popularity Lift} = \frac{\frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|L_q|} \sum_{v \in L_q} d(v)}{\frac{1}{|C|} \sum_{v \in C} d(v)} \tag{14}$$

This is a graph-degree adaptation of popularity lift in recommendation-bias evaluation [7]. Values above 1 indicate that a model tends to recommend higher-degree articles than the candidate-pool baseline.

Aggregate Diversity counts how many distinct articles appear at least once across all recommendation lists [10, 18]:

$$\text{Aggregate Diversity} = \left| \bigcup_{q \in Q} L_q \right| \tag{15}$$

Novelty is measured with a graph-degree-based self-information score,

$$\text{Novelty} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|L_q|} \sum_{v \in L_q} -\log_2 \left( \frac{d(v)+1}{M} \right) \tag{16}$$

which is the graph-adapted form used in our implementation rather than an interaction-frequency definition [18]. For Intra-List Diversity (ILD), we average pairwise cosine distances within each recommendation list in semantic embedding space:

$$\text{ILD} = \frac{1}{|Q|} \sum_{q \in Q} \frac{2}{|L_q|(|L_q|-1)} \sum_{i<j,\ v_i, v_j \in L_q} (1 - \cos(e(v_i), e(v_j))) \tag{17}$$

where $e(v)$ denotes the BGE-M3 embedding of article $v$ and the formula is applied to lists with at least two items [18]. Finally, let $c_{(1)} \leq \cdots \leq c_{(n)}$ denote the sorted recommendation counts of the distinct articles that appear in at least one list. The Gini Index is

$$\text{Gini} = \frac{\sum_{i=1}^{n} (2i - n - 1) c_{(i)}}{n \sum_{i=1}^{n} c_{(i)}} \tag{18}$$

which summarizes how unevenly recommendation exposure is distributed over the catalog [18]. Higher values are better for Hit-based metrics, MRR, NDCG, MAP, Aggregate Diversity, Novelty, ILD, ROC-AUC, and PR-AUC, whereas lower values are better for Popularity Lift and Gini Index.

### 4.2 Results

**Combined Metrics on the Revised Protocol** Table 2 summarizes the main revised evaluation results. The corrected hybrid family occupies the strongest overall positions in top-list ranking metrics, while structural-only models remain competitive on some recommendation-quality indicators.

**Table 2.** Combined ranking and recommendation-quality metrics under the revised evaluation protocol

| Model | H@1 | H@10 | H@50 | H@100 | MRR | NDCG | Recall | MAP | Pop. Lift | Gini | Agg. Div. | Novelty | ILD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGE pooled + Node2Vec (corrected) | 1.03% | 8.67% | 31.46% | 47.81% | 0.0410 | 0.1160 | 0.4781 | 0.0392 | 1.3301 | 0.4110 | 11,326 | 14.2738 | 0.4372 |
| BGE-M3 + Node2Vec (corrected) | 1.07% | 8.67% | 32.41% | 49.81% | 0.0421 | 0.1202 | 0.4981 | 0.0403 | 1.3081 | 0.4127 | 11,268 | 14.2783 | 0.4316 |
| BGE pooled + Node2Vec (uncorrected) | 0.76% | 7.83% | 32.64% | 50.74% | 0.0376 | 0.1176 | 0.5074 | 0.0358 | 1.3397 | 0.4613 | 11,186 | 14.0351 | 0.4558 |
| BGE-M3 + Node2Vec (uncorrected) | 0.76% | 7.83% | 32.64% | 50.74% | 0.0376 | 0.1176 | 0.5074 | 0.0358 | 1.3400 | 0.4605 | 11,164 | 14.0340 | 0.4557 |
| Node2Vec (uncorrected) | 0.75% | 7.81% | 32.52% | 50.60% | 0.0375 | 0.1172 | 0.5060 | 0.0357 | 1.3374 | 0.4482 | 10,966 | 14.0466 | 0.4569 |
| Node2Vec (corrected) | 0.75% | 7.81% | 32.52% | 50.60% | 0.0375 | 0.1172 | 0.5060 | 0.0356 | 1.3374 | 0.4482 | 10,966 | 14.0466 | 0.4569 |
| GCN + BGE-M3 (rank fusion) | 0.85% | 7.63% | 23.14% | 33.77% | 0.0335 | 0.0862 | 0.3377 | 0.0317 | 1.6650 | 0.4228 | 11,655 | 13.9222 | 0.4366 |
| BGE-M3 Only | 1.35% | 7.57% | 19.69% | 27.73% | 0.0366 | 0.0788 | 0.2773 | 0.0351 | 1.9413 | 0.5172 | 10,905 | 13.8459 | 0.3713 |
| GCN Only | 0.38% | 3.46% | 14.12% | 22.91% | 0.0180 | 0.0530 | 0.2291 | 0.0163 | 1.1879 | 0.2607 | 11,685 | 14.3293 | 0.5135 |
| GraphSAGE Only | 0.17% | 1.22% | 4.96% | 8.72% | 0.0076 | 0.0199 | 0.0872 | 0.0061 | 1.3134 | 0.2517 | 11,628 | 14.2087 | 0.5313 |

Two patterns are visible in Table 2. First, the corrected hybrid variants occupy the top positions on Hit@10 and maintain strong placements on MRR, NDCG, Recall, and MAP. Second, the model ranking is not identical across all recommendation-quality measures. Structural-only models, for example, remain competitive on some novelty and ILD statistics, while semantic-only retrieval shows different popularity behavior.

**Cutoff-10 and Link-Discrimination Metrics** Table 3 adds a smaller-cutoff view together with ROC-AUC and PR-AUC. This table helps separate recommendation behavior near the top of the list from broad discrimination across positive and negative edges.

**Table 3.** Cutoff-10 ranking metrics and link-discrimination metrics on the revised protocol

| Model | H@10 | NDCG@10 | Recall@10 | MAP@10 | ROC-AUC | PR-AUC |
|---|---|---|---|---|---|---|
| BGE pooled + Node2Vec (corrected) | 8.67% | 0.0409 | 8.67% | 0.0273 | 0.9563 | 0.9559 |
| BGE-M3 + Node2Vec (corrected) | 8.67% | 0.0413 | 8.67% | 0.0278 | 0.9582 | 0.9579 |
| BGE pooled + Node2Vec (uncorrected) | 7.83% | 0.0356 | 7.83% | 0.0229 | 0.9710 | 0.9716 |
| BGE-M3 + Node2Vec (uncorrected) | 7.83% | 0.0356 | 7.83% | 0.0229 | 0.9710 | 0.9716 |
| Node2Vec (uncorrected) | 7.81% | 0.0355 | 7.81% | 0.0229 | 0.9717 | 0.9714 |
| Node2Vec (corrected) | 7.81% | 0.0355 | 7.81% | 0.0228 | 0.9717 | 0.9714 |
| BGE-M3 Only | 7.64% | 0.0406 | 7.64% | 0.0298 | 0.8851 | 0.8886 |
| GCN + BGE-M3 (rank fusion) | 7.63% | 0.0355 | 7.63% | 0.0234 | 0.9356 | 0.9363 |
| GCN Only | 3.46% | 0.0161 | 3.46% | 0.0107 | 0.8954 | 0.8946 |
| GraphSAGE Only | 1.22% | 0.0060 | 1.22% | 0.0041 | 0.8691 | 0.8561 |

The cutoff-10 view refines the interpretation of the benchmark. The corrected hybrid variants remain strongest on Hit@10, while the non-pooled corrected hybrid is highest on NDCG@10 among the compared models. At the same time, BGE-M3 Only remains competitive on MAP@10. On the broader discrimination side, uncorrected structural and lightly fused models occupy the highest ROC-AUC and PR-AUC values. This indicates that the model family that works best for top-ranked recommendation is not necessarily the same one that works best for global edge discrimination.

**Discussion** Taken together, the revised evaluation suggests that the corrected hybrid family offers the most balanced overall recommendation behavior for the present task. The pooled corrected hybrid reaches the top Hit@10 value, while the non-pooled corrected hybrid remains strong across MRR, NDCG, Recall, and several broader quality measures. These models benefit from access to both text-level semantic proximity and graph-level structural context, without relying exclusively on either.

Two design choices deserve separate interpretation. First, pooling was introduced to balance the dimensional contribution of the 1024-dimensional semantic vector and the 128-dimensional structural vector. In the revised results, that balancing effect is visible but modest: the pooled corrected hybrid ties for the best Hit@10, whereas the non-pooled corrected hybrid remains slightly stronger on MRR, NDCG, Recall, and MAP. Pooling is therefore useful as an alignment device, but it does not by itself explain the full gain of the hybrid family.

Second, degree correction behaves differently depending on where it is used. In pure Node2Vec retrieval, the corrected and uncorrected variants are nearly identical in the revised tables. In the hybrid setting, however, the corrected variants improve substantially over their uncorrected counterparts on top-list metrics such as Hit@10 and MRR. This suggests that the heuristic matters less as an isolated structural adjustment than as a way of stabilizing the interaction between semantic and structural signals inside the hybrid representation.

The structural-only models remain informative, particularly when the evaluation is interpreted as link discrimination rather than recommendation at the top of a ranked list. Conversely, the semantic-only setting remains competitive on some precision-sensitive metrics while showing a different popularity profile. This split supports a practical conclusion: model selection depends on whether the main objective is broad edge separation or short ranked recommendation lists.

The benchmark layer also matters. Because the revised protocol is aligned with the undirected structural view, it provides a cleaner basis for comparing semantic and structural retrieval families on the same task. In that sense, the reconstructed dataset and the revised evaluation protocol are not only supporting infrastructure; they are part of what makes the comparison interpretable.

## 5 Conclusion

This paper presented a graph-based article-to-article recommendation framework for Wikipedia built around semantic, structural, hybrid, and GNN-based retrieval families. The work also introduced Custom-WikiCS, a reconstructed and text-complete version of Wiki-CS designed to support modern recommendation experiments with Transformer-based semantic representations.

Within this benchmark, the compared models showed different strengths. The revised evaluation indicates that corrected semantic-structural hybrids occupy the strongest overall position for recommendation-oriented top-list retrieval, while structural-only models remain competitive under global discrimination metrics such as ROC-AUC and PR-AUC. This suggests that hybrid retrieval is a useful direction when the target task is ranked article recommendation rather than graph prediction in isolation.

Future work can extend this line in several directions, including broader Wikipedia domains, alternative structural correction schemes, and richer hybrid fusion strategies that preserve interpretability while improving recommendation quality.

**Availability of Data and Code** The codebase and dataset reconstruction artifacts used in this study are currently available at `https://github.com/regsah/graph-final-project`.

**Disclosure of Interests.** The authors have no competing interests to declare.

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